Exercises

一、Sequences

1. Given the sequences of cities:

   $u, v : \mathrm{seq}\ \mathrm{CITY}$

   and the values

   $u = \langle \mathrm{London}, \mathrm{Amsterdam}, \mathrm{Madrid} \rangle$

   and

   $v = \langle \mathrm{Paris}, \mathrm{Frankfurt} \rangle$

   write down the values of the sequences:

   $u \frown v$
   $\mathrm{rev}\ (u \frown v)$
   $\mathrm{rev}\ u$
   $\mathrm{rev}\ v$
   $\mathrm{rev}\ v \frown \mathrm{rev}\ u$

2. Referring to Question 1, find the value of

   $\mathrm{squash}\ (2 \mathinner{..} 4 \lhd \mathrm{rev}\ (u \frown v))$

3. Find the value of

   $\mathrm{squash}\ (4 \mathinner{..} 2 \lhd \mathrm{rev}\ (u \frown v))$

4. Find the value of

   $u \frown v \upharpoonright \{ \mathrm{London}, \mathrm{Moscow}, \mathrm{Paris}, \mathrm{Rome} \}$

5. Find the value of

   $\mathrm{tail}\ (u \frown v) \frown \mathrm{front}\ \langle \mathrm{Moscow}, \mathrm{Berlin}, \mathrm{Warsaw} \rangle$

二、Specifications

1. 设 Instruments 为乐器的给定集合，Actions 为对乐器施行的动作的集合，并有如下两个定义在给定集合People、Instruments 和 Actions 上的关系。

plays = = {(LiGang $\mapsto$ piano) , (NiJun $\mapsto$ guitar), (DaWei $\mapsto$ violin),
(ZhangHua $\mapsto$ trumper), (ZhangYing $\mapsto$ flute), (ZhangYing $\mapsto$ piano),
(GaoLei $\mapsto$ piano)}

worksBy = = {(piano $\mapsto$ hammering) , (guitar $\mapsto$ plucking), (harpsichord $\mapsto$ plucking),
(trumper $\mapsto$ blowing), (flute $\mapsto$ blowing), (violin $\mapsto$ bowing),
(violin $\mapsto$ scraping)}

(1) 分别写出 plays 和worksBy 的domain 和 range

(2) 写出它们的类型

(3) 写出 play~ 以及它的domain 和 range

(4) 写出 worksBy~ 以及它的domain 和 range

2. 用上题中定义的关系 plays 和 worksBy 写出如下定义的每一个关系的内容。

(1){piano , harpsichord } $\lhd$ worksBy

(2) plays $\rhd$ {piano , violin}

(3) plays $\rhd$ {piano}

(4) worksBy $\rhd$ {bowing , scraping}